for Quadruped,

define: P_{Bi} = foot position in body frame.

P_{Li} = foot position in leg frame.

P_{wi} = foot position in world frame.

R_{wi} = body pose in world.

$$\left(\frac{dP_{wi}}{dt}\right) = \frac{d(P_{Bi})}{dt} + \omega_{IMU} \times P_{Bi}$$

$$P_{wi} = B_{wi} + R_{WB} \cdot P_{Bi} \rightarrow \text{foot posit body.}$$

P_{wi}: foot posi world

B_{wi}: Body posi world

R_{WB}: rotational matrix of Body relative to world.

$$\frac{dP_{wi}}{dt} = \left(\frac{dB_{wi}}{dt}\right) + \frac{d(R_{WB} \cdot P_{Bi})}{dt}.$$

$$V_{wi} = V_{Bwi} + \dot{R}_{WB} P_{Bi} + R_{WB} \cdot \dot{P}_{Bi}$$

$$= V_{Bwi} + R_{WB}(\omega_{IMU} \times P_{Bi}) + R_{WB} \cdot \dot{P}_{Bi}$$

$$V_{wi} = V_{Bwi} + R_{WB}\left(\dot{P}_{Bi} + \omega_{IMU} \times P_{Bi}\right)$$

$\dot{P}_{Bi}$: foot pos in body ↑ = foot vel in body. ↓

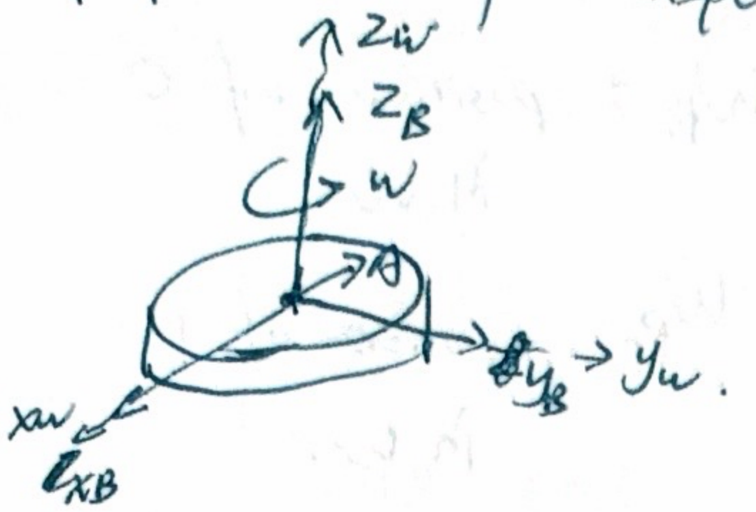
V_{wi}: foot velocity in world

V_{Bwi}: body velocity in body world. — body translation

R_{WB}: rotational matrix that convert from body to world.

ω_{IMU}: IMU velocity

Full Derivation of transport theorem:



$$\vec{A}_B = \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}.$$

$$R_{BW} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$\vec{w} = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix}, \quad \text{so} \quad \vec{A}_W = \begin{bmatrix} A_x\cos\theta - A_y\sin\theta \\ A_x\sin\theta + A_y\cos\theta \\ A_z \end{bmatrix}$$ (a position vector in world)

find velocity of $\vec{A}$:

$$\dot{R}_{BW} = \begin{bmatrix} -\sin\theta\cdot\dot{\theta} & -\cos\theta\,\dot{\theta} & 0 \\ \cos\theta\cdot\dot{\theta} & -\sin\theta\,\dot{\theta} & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$\left(\frac{d\vec{A}_W}{dt}\right) = \begin{bmatrix} -A_x\sin\theta\cdot\dot{\theta} - A_y\cos\theta\cdot\dot{\theta} \\ A_x\cos\theta\cdot\dot{\theta} - A_y\sin\theta\cdot\dot{\theta} \\ 0 \end{bmatrix}.$$

since $\vec{w} \times \vec{A}_W = \begin{vmatrix} i & j & k \\ 0 & 0 & \dot{\theta} \\ A_x\cos\theta - A_y\sin\theta & A_x\sin\theta + A_y\cos\theta & A_z \end{vmatrix}$

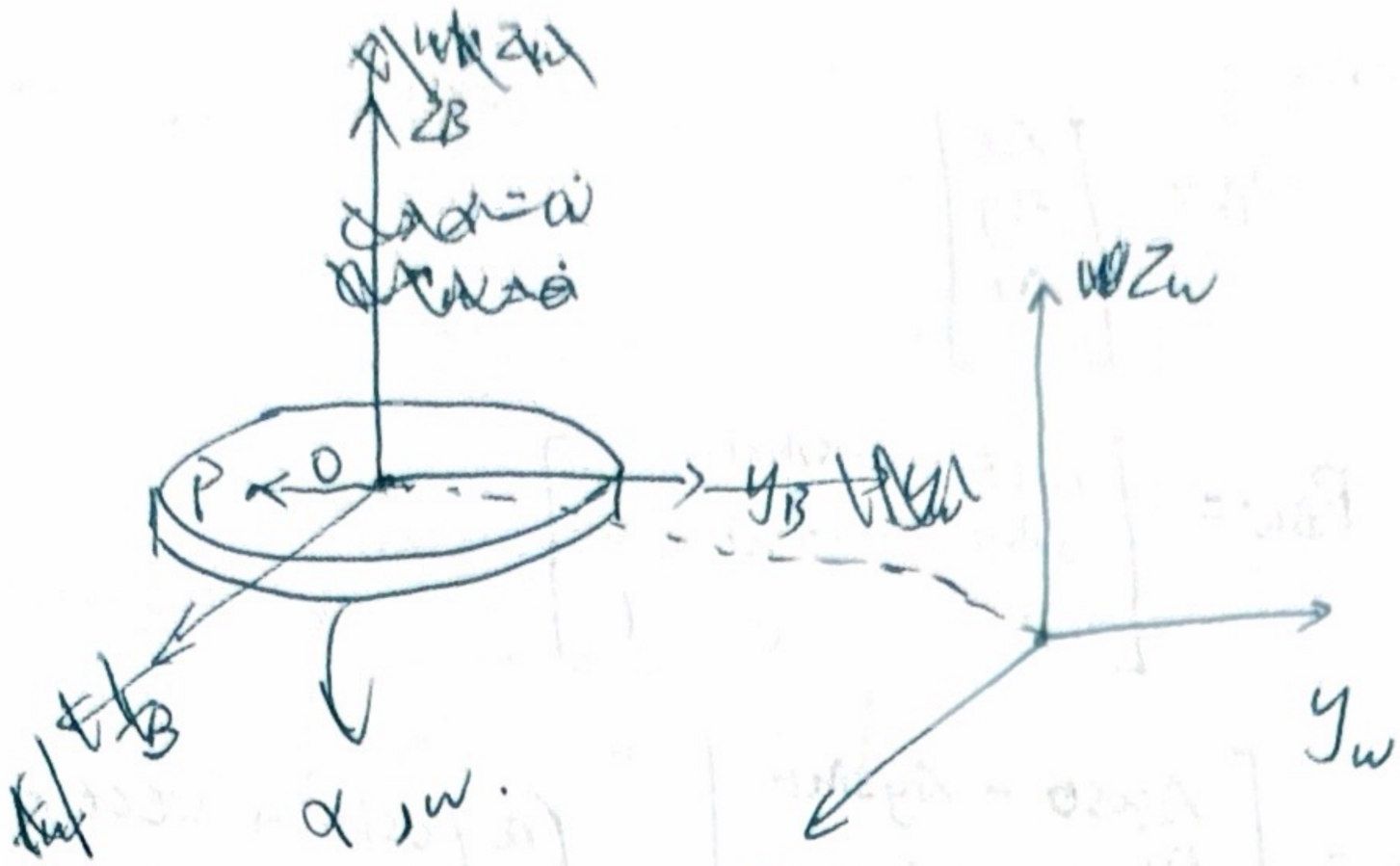
so $\dot{R}_{BW}\cdot\vec{A}_B$

$= R_{BW}\cdot(\vec{w}\times\vec{A}_B)$ $= \begin{bmatrix} -A_x\sin\theta\cdot\dot{\theta} - A_y\cdot\dot{\theta}\cos\theta \\ A_x\dot{\theta}\cos\theta - A_y\cdot\dot{\theta}\sin\theta \\ 0 \end{bmatrix} = \left(\frac{d\vec{A}_W}{dt}\right)$

so $\frac{d\vec{A}_W}{dt} = \vec{w}\times\vec{A}_W$ (if A_x, A_y fixed).

generally, since $\vec{A}_W = R_{BW}\cdot\vec{A}_B$

so $$\frac{d\vec{A}_W}{dt} = \dot{R}_{BW}\cdot\vec{A}_B + R_{BW}\cdot\dot{\vec{A}}_B$$
$$= R_{BW}(\vec{w}\cdot\vec{A}_W\times\vec{A}_B) + \left(\frac{d\vec{A}_B}{dt}\right)R_{BW}$$



W_{P0} = position of O in W.

W_{PP} = position of P in W.

$r_{P/0}$ = $\vec{OP}$ in O frame.

$$W_{PP} = (r_{P/0})_W + W_{P0}$$

$$W_{VP} = W_{V0} + \left(\frac{d(r_{P/0})}{dt}\right)_W$$

$$\left(\frac{dw_{PP}}{dt}\right) = \left(\frac{d\,r_{P/0}}{dt}\right)_W + \left(\frac{dw_{P0}}{dt}\right) \Rightarrow$$

since transport theorem:

$$\left(\frac{dr}{dt}\right)_W = \left(\frac{dr}{dt}\right)_0 + w \times r. \quad \text{, let } \left(\frac{dr}{dt}\right)_0 = W_{rel\,r}$$

so $W_{VP} = W_{V0} + W_{V\,rel\,r} + w \times r.$

$$\left(\frac{dW_{VP}}{dt}\right) = \frac{dW_{V0}}{dt} + \left(\frac{d\,W_{V\,rel\,r}}{dt}\right)_W + \frac{d(w \times r)}{dt}$$

$$W_{\alpha P} = W_{\alpha 0} +$$

$$\left(\frac{dr}{dt}\right)_W = \left(\frac{dr}{dt}\right)_0 + w \times r$$

$$= V_{rel} + w \times r$$

① $\left(\frac{d\,W_{rel\,r}}{dt}\right)_W = \left(\frac{d\,W_{rel\,r}}{dt}\right)_0 + w \times V_{rel}$

② $\frac{d(w \times r)}{dt} = \dot{w} \times r + w \times \dot{r} = \alpha \times r + w \times (V_{rel} + w \times r)$

so $\alpha_P = \underbrace{\alpha_0}_{\text{body O}} + \underbrace{\alpha_{rel}}_{\text{point in O}} + \underbrace{2w \times V_{rel}}_{\text{Coriolis}} + \underbrace{w \times (w \times r)}_{\text{centripetal.}} + \underbrace{\alpha \times r}_{\text{tangential.}}$ (in W)

If ignore body translational:

$$V_{wi} = R_{wB} \cdot \left(\dot{P}_{Bi} + W_{Imu} \times P_{Bi} \right). \quad \text{(world frame)}$$

$$R_{BwB} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{yaw.}$$

$$R_{wB} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \quad \text{pitch}$$

$$R_{wB} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \quad \text{roll.}$$

Simple IMU Correction PD Controller:

$$B_{err} = \begin{bmatrix} \phi_{desired} - \phi_i \\ \theta_{desired} - \theta \\ 0 \end{bmatrix} \quad \begin{matrix} \text{pitch} \\ \text{roll} \\ \text{yaw.} \end{matrix}$$

$B_w = B\dot{\theta}$ (IMU angular velocity)

$$B_{wcorr} = K_p \cdot B_{err} - K_w \cdot B_w$$

$$B_{pi} = B_{wcorr} \times B_{pi}$$

B_{pi}: foot velocity in body frame

B_{wcorr}: IMU angular velocity

B_{pi}: foot position in Body frame

since foot position body frame ↓ pure translational foot position leg frame

so $B_{pi} = p_i$ (leg frame).